

Overview:

This documentation outlines the step-by-step process, technologies utilised, and challenges encountered during the development of the web application. The application is designed to provide a user-friendly dashboard interface for managing data and accessing various functionalities.

1. Project Setup:

- Tech Stack - The web app was built using React for the frontend. React Router was employed for client-side routing and navigation.
- Initial Setup - The project was initiated using Create React App, providing a basic scaffold, allowing us to develop.
- File Structure: The project structure followed a modular pattern, dividing components, styles, and utilities into organised directories for better maintainability.

2. Components:

- DashboardComponent: This component constitutes the main dashboard view. It comprises widgets displaying various data graphs and information related to energy consumption, carbon neutrality, CO2 emissions, and project details.
- LoginComponent: Responsible for user authentication, the LoginComponent contains a form for entering credentials. Upon successful login, users are redirected to the dashboard.
- ProtectedRouteComponent: Implemented to protect routes based on authentication status. If a user is not logged in, they are redirected to the login screen.
- AccountPage Component - The core component responsible for rendering the account settings interface. It has Segmented sections for managing user information, including profile details and various account settings options.
- DataManager Component - This is the main page for displaying the user's data. They will be able to access pages allowing the user to update, delete data and add data stored in a database.

3. Routing and Navigation:

- React Router: React Router facilitated navigation within the app. Routes were defined to display different components based on the URL path.
- Protected Routes: The ProtectedRouteComponent leveraged React Router to ensure certain routes were accessible only if the user was logged in.

4. Styling and UI:

- Styling Approach: Inline styles were initially used for basic styling. Later, a CSS-in-JS solution like Styled Components or CSS modules could be considered for better maintainability and scoping of styles.
- Responsive Design: The app was designed to be responsive, adapting to various screen sizes and orientations for a consistent user experience.

5. Challenges Faced:

- Authentication Flow: Implementing a robust authentication flow and handling user sessions securely posed challenges, particularly managing authentication state across components.
- Styling Consistency: Maintaining consistent styling across different components and ensuring UI uniformity was a challenge, especially as the app grew in complexity.
- Performance Optimisation: Initial concerns about performance optimisation arose as the app's data visualisation components were resource-intensive.

Testing Table:

Component/File	Functionality Tested	Testing Method	Expected Outcome	Test Result
LoadingScreenComponent.js	Rendering loading screen	Visual inspection	Loading screen displayed with Eviden logo, slogan and team image	Pass
	Navigation to login on click	Click simulation	Redirects to the '/login' page	Pass
LoginComponent.js	Rendering login form	Visual inspection	Login form with fields for username, password, and buttons	Pass
	Login functionality	Simulate valid login	Successful login redirects to '/dashboard'	Pass

	Login functionality	Simulate invalid login	Invalid alert pops up saying the invalid user or password was inputted, reinput	Pass
	Sign up and Forgot Password buttons functionality	Click simulation	-	Not Tested
DashboardComponent.js	Rendering of dashboard components and navigation	Visual inspection	Dashboard displayed with navigation links and content	Pass
	Log out functionality	Click simulation	Removes login status and redirects to '/'	Pass
AccountPage.js	Rendering of account settings components and navigation	Visual inspection	Account settings displayed with navigation links and content	Pass

	Change Name, Email, Password, Profile functionality	Click simulation	-	Not Tested
App.js	Routing setup	Inspection of routes and navigation	Routes '/login', '/dashboard', '/account' working correctly	Pass
ProtectedRouteComponent.js	Handling protected routes based on login status	Inspection of routing logic	'/dashboard' accessible when logged in, else redirects to '/login'	Pass